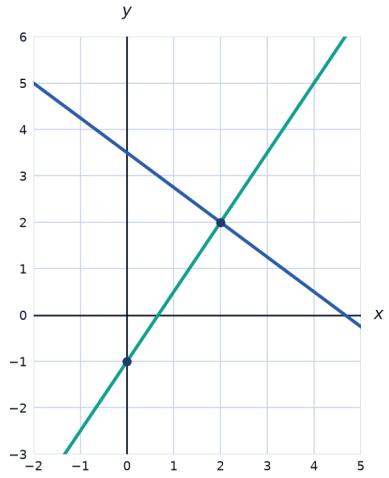
Powers of whole numbers

Core Concepts: Powers of whole numbers

1 For the model $f(x) = 6x + 5$ in powers of whole numbers, evaluate $f(2)$.



a. 17

b. 23

c. 12

d. 21

2 Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

3 A data set recorded for powers of whole numbers consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 6x + 5$ in powers of whole numbers, evaluate $f(3)$.

a. 23

b. 29

c. 18

d. 27

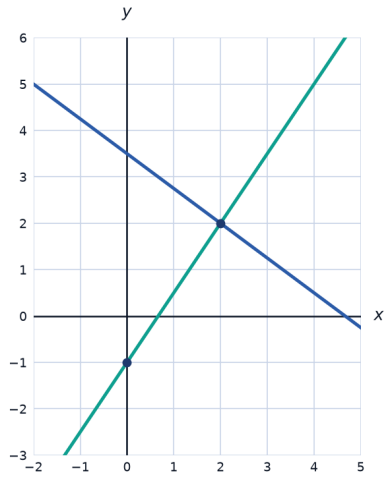
5 Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$



6 A data set recorded for powers of whole numbers consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

Graphical & Geometric Interpretation

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7 For the model $f(x) = 6x + 5$ in powers of whole numbers, evaluate $f(4)$.

a. 29

b. 35

c. 24

d. 33

8 Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

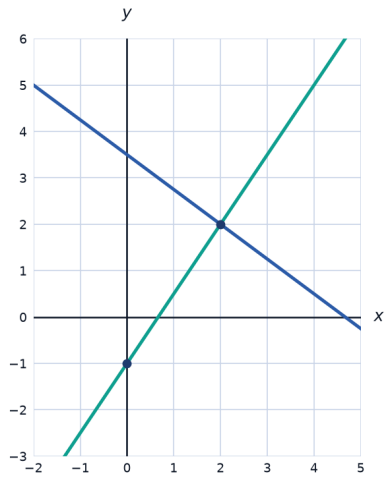
a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

9 A data set recorded for powers of whole numbers consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?



a. 14.5

b. 15.5

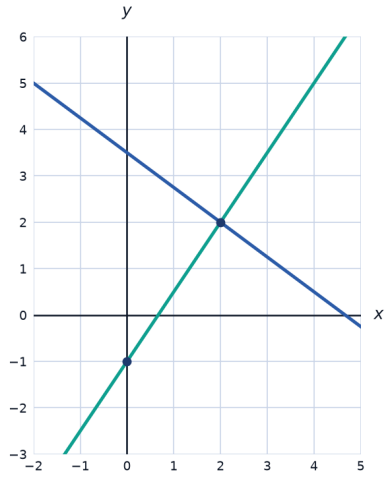
c. 16.5

d. 18.5

Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 6x + 5$ in powers of whole numbers, evaluate $f(5)$.



a. 35

b. 41

c. 30

d. 39

11 Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

12 A data set recorded for powers of whole numbers consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

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